The Aglet Merchant
Problem ID: TheAgletMerchant

Unfortunately, you work for an aglet merchant who has a vast fleet of caravans to transport his wares across a desert. Recently, a group of robbers has been plundering his caravans. As his lowly servant, he commands you to redesign the distribution process to ensure that he stays filthy rich. As the slimy, greedy pessimist he is, the merchant wants you to maximize how much he is guaranteed to earn even if the robbers attack the worst possible caravan. You know that the robbers only have enough men to rob one caravan.

Trading posts are just places where caravans can go to and deliver the wares to the next caravans, but opposed to cities, the aglets never end there, they just pass through.

Since the robbers might not attack, and the merchant does not want to lose profit, he will always send as many aglets as he can, limited only by the caravan's capacity and the Trade Federation's imposed limits.

Input

The input consists of:

- One line containing two numbers, C and P, where C is the price to produce an aglet, and P is the selling price of an aglet.
- A second line contains three numbers, N, the number of cities he exports aglets to, T, the number of trading posts that the caravans can pass through, R, the number of caravans that go between trading posts and cities. The cities are numbered 1 to N, and the trading posts are numbered N+1 to N+T. The merchants ships all his aglets from his warehouse at 0.
- Then follows N lines, each containing an integer which is the amount of aglets The Trade Fedaration has allowed you to import into the N'th city.
- Lastly followes by R lines, each containing 3 integers. The first integer is the starting point, the second is the endpoint, and the third is the amount of aglets the caravan can carry. There can only be one caravan between each a given starting- and ending point.

Output

If the merchant can make money on the trade route, output two integers, the maximum number of aglets that reaches the cities if the robber attacks the worst possible caravan, and the amount of money he makes if he sells them all. If he can not make money, output "Not worth".